STEAMROOT · TRACEABILITY MATRIX

SteamRoot traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/steamroot		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

SteamRoot traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Pasteurize soil to working depth	STR-CAL-001, STR-BLD-001, STR-STM-001, STR-OPT-001	Analysis (CAL-001 section 2), TRL 3; thermocouple probes at 15 cm, TRL 4	To verify at TRL 4 (1D model holds 70 °C at 15 cm if the front reaches 16.5 cm; STEM S11 fails: in soils finer than sand the steam cannot pass at full flow under the hood; bed preparation and hold-down proposed, DEC #18)
UR-002	PR-002	Treatment rate at 15 cm	STR-CAL-001, STR-STM-001	Analysis (CAL-001 section 3), TRL 3; timed trial, TRL 4	At risk: 1.88 m2/h; about 1.5 to 1.6 m2/h net of edge strips, overlap and hose losses (STEM M2)
UR-003	PR-003	Steam output	STR-CAL-001, STR-BLD-001	Analysis (CAL-001 section 4), TRL 3; feed metering, TRL 4	Met on paper: 30 kg/h at 32.2 kW firing
UR-004	PR-004	Wood per m2	STR-CAL-001	Analysis (CAL-001 section 4), TRL 3; fuel weighing, TRL 4	Not met: 4.36 kg/m2 with the baffles and dampers set (4.44 with the dampers wide open); restatement to 4.5 kg/m2 proposed, awaiting Amish (DEC #12)
UR-004	PR-005	Fuel-to-steam efficiency	STR-CAL-001, STR-BLD-001, STR-STM-001	Analysis (CAL-001 section 4), TRL 3; flue temperature, oxygen and fuel weighing, TRL 4	At risk: 66.1 % at λ = 1.5 with the gas-pass baffles, a margin smaller than its uncertainties (62.5 to 66.9 % in CAL Table 6a; 59.8 % without baffles)
UR-005	PR-006	Header pressure and open vent	STR-CAL-001, STR-BLD-001, STR-STM-001	Analysis (CAL-001 section 5); design review, TRL 3	Met on paper: 0.033 bar; seal limit 0.084 bar; vent cannot be isolated
UR-005	PR-007	Stored water	STR-CAL-001, STR-BLD-001, STR-STM-001	Analysis (CAL-001 section 6), TRL 3	Met on paper: 7.3 L flooded
UR-006	PR-008	Run between refills	STR-CAL-001, STR-BLD-001	Analysis (CAL-001 section 7), TRL 3; timed run, TRL 4	At risk: water 4.2 h; the 20 Ah battery runs to about 100 % depth of discharge per fill at an assumed 4 A pump (STEM E3; DEC #16)
UR-005	PR-009	Certified relief valve	STR-CAL-001, STR-BLD-001	Valve rating and capacity check (CAL-001 section 6), TRL 3	Met on paper: rated 170 kg/h against a 141 kg/h flash on a 30 s basis; the flash basis is open (STEM S16, DEC #20); confirm on the valve bought
UR-007, UR-013	PR-010	Operator and site protection	STR-CAL-001, STR-	Design review, TRL 3; surface	

User requirement	Product requirement	Title	Evidence	Verification	Status
			BLD-001, STR-STM-001	temperature check, TRL 4	Met by design: 3.2 m (raised for draft, DEC S3), 6 mm mesh, hood skin 23 °C
UR-008	PR-011	Towed mass	STR-CAL-001, STR-BLD-001, STR-STM-001, BOM	Analysis (CAL-001 section 7 and the model), TRL 3; weighbridge, TRL 4	Not met: 509 kg with both hoods aboard (643 kg full; 447 kg without the hoods, which ride on a second vehicle for the prototype; basis proposed in DEC #13)
UR-009	PR-012	Parts cost	STR-CAL-001	Priced BOM, TRL 3	Over target: USD 2,720 (USD 520 over; the overrun to USD 2,600 was accepted by Amish on 2026-10-03, the USD 120 of STEM fixes is new)
UR-005	PR-013	Coil inlet pressure	STR-CAL-001, STR-BLD-001, STR-STM-001	Analysis (CAL-001 section 5), TRL 3	Met on paper: 0.129 bar
UR-002	PR-014	Treatment rate at 5 cm	STR-CAL-001	Analysis (CAL-001 section 3), TRL 3	At risk: 3.41 m2/h
UR-008	PR-015	Overall width	STR-CAL-001, STR-BLD-001, STR-STM-001, BOM	Model and drawing STR-DWG-002, TRL 3	At risk: 1.48 m
UR-010	PR-016	Hood handling	STR-CAL-001, STR-BLD-001, STR-STM-001	Analysis (STEM B4), TRL 3; handling trial, TRL 4	Not met for one person: 31.3 kg hood, handles 1.12 m apart; a two-person share of 15.6 kg is borderline once the reach is counted. Wheel kit recommended; proposed, awaiting Amish (DEC S2)
UR-011	PR-017	Two independent means against pressure	STR-CAL-001, STR-STM-001	Design review, TRL 3; written ruling from TDLR before any build	Met by design (water seal and certified relief valve); ruling not yet obtained. Proposed, awaiting Amish (DEC #17)
UR-012, UR-014	PR-018	Workshop build	STR-CAL-001, STR-BLD-001	Inspection of the model, BOM and build plan, TRL 3	Met by design (STR-BLD-001; `cad/fidelity.json` D3). Proposed, awaiting Amish (DEC #17)

Gaps

None.